

Exercise 01 SQL Fundamentals: Select & Filters

1. SELECTING & SORTING

① Retrieve all columns from the employees table.

Or
SELECT *
FROM employees;
Employees

ID	First name	Last name	Dep	Salary	Hire date	City
1	Alice	Green	IT	70000	20-01-10	Johannesburg
2	Brian	Lee	HR	45000	19-03-22	Cape Town
3	Carol	Zulu	Finance	65000	18-07-12	Durban
4	David	Makoni	Marketing	50000	21-11-05	Pretoria
5	Eva	Kwanda	IT	72000	17-09-20	Johannesburg

② Find all unique departments:

Select department

Department
IT
HR
Finance
Marketing

③ Retrieve first and last names ordered by salary descending

SELECT first name, last name
FROM employees
ORDER BY Salary Desc;

first name	last name
Guy	Naruto
Mike	Green
Conny	Zulu
Davis	Wilson

④ Retrieve the top 3 highest-paid employees

SELECT (1), first name, last name
FROM employees
ORDER BY Salary Desc;
LIMIT 3;

ID	first Name	last Name
5	Guy	Naruto
1	Mike	Green
3	Conny	Zulu

2. Functions with Where

⑤ find employees in the IT department

SELECT ID, first_name, last_name
FROM employees
WHERE department = "IT";

ID	First name	Last Name	Department
1	Alice	Green	IT
5	Gee	Naidu	IT

⑥ find employees in finance with Salary > 60000

SELECT ID, first_name
FROM employees
WHERE department = "finance"
AND Salary > 60000

ID	First name	Last Name	Department	Salary
3	Colby	Turner	finance	65000

⑦ find employees in HR or Marketing

SELECT ID
FROM employees
WHERE department = "HR"
OR department = "Marketing";

ID	First name	Last name	Department
2	Brian	Lee	HR
4	David	Molnar	Marketing

⑦ find employees in HR not in IT.

Select id, first name, last name, department
from employees
where department != "IT";

ID	First name	Last name	Department
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
4	David	Molnar	Marketing

⑧ find employees in IT, HR, or finance using in.

Select id, first name, last name, department
from employees
where department in ('IT', 'HR', 'Finance');

ID	First name	Last name	Department
1	Alice	Green	IT
5	Chris	Nazario	IT
2	Brian	Lee	HR
3	Cathy	Zulu	Finance

⑩ find in employees in IT with Salary > 65000 and City Johannesburg

Select ID, first name, last name
 from employees
 where Department = 'IT'
 and Salary > 65000
 and City = 'Plymouth';

ID	First Name	Last Name	Dept	Salary	City
1	Mike	Green	IT	70000	Tokyo
5	Kim	DeVere	IT	72000	Plymouth